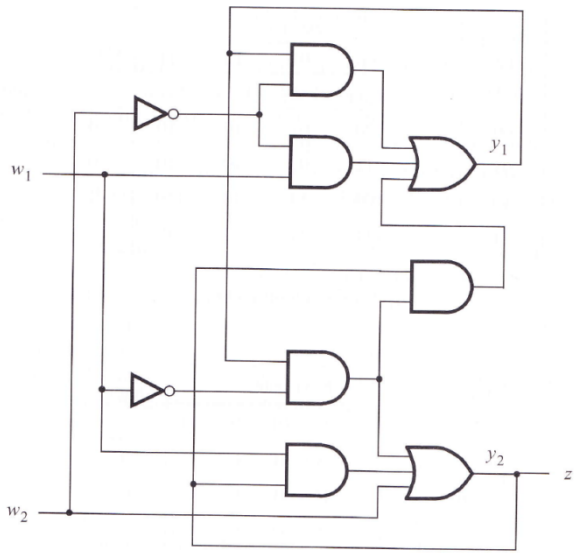
CE2003: Tutorial 10

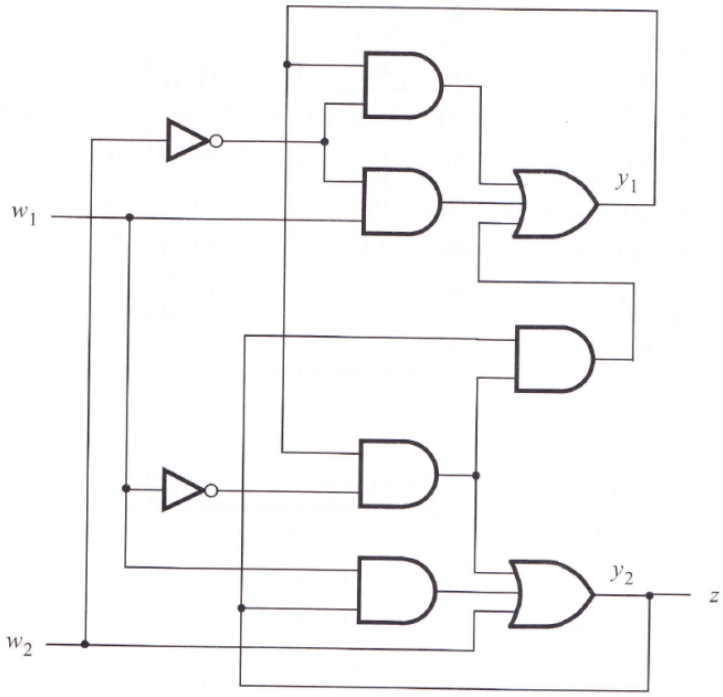
Q1 Determine if there are any hazards in the circuit below. (Same circuit as in Q2 of Tut 9).



Q2: Determine the circuit for the following state flow diagram

Present State	$w_2 w_1$ 00	01	10	11	Output z
A	**A**	E	C	--	0
B	--	E	G	**B**	1
C	A	--	**C**	F	0
D	A	**D**	--	B	1
E	A	**E**	--	B	0
F	--	D	C	**F**	0
G	A	--	**G**	B	1

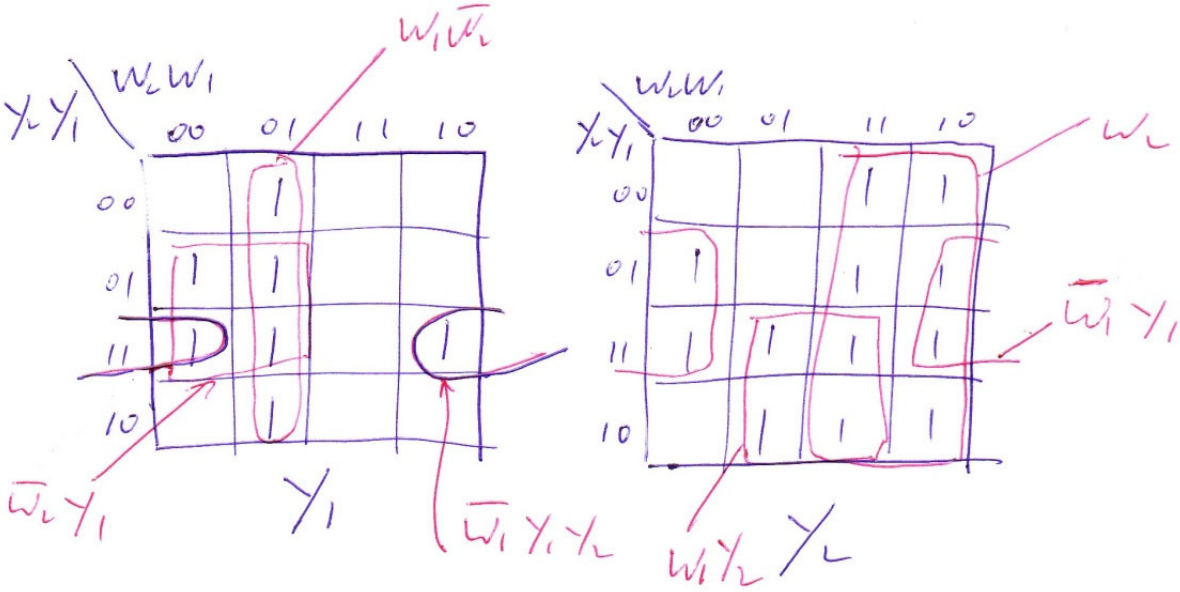
Answers: Q1: To determine if there are hazards, we need to do two things. First examine the K-map and make sure ALL prime implicants are covered, this will eliminate static hazards. Second, we need to check the state transition diagram to make sure there are no diagonal transitions as these could result in dynamic hazards.



Next state & output expressions (from previous tut) are:

$$Y_1 = w_1 w_2' + w_2' y_1 + w_1' y_1 y_2$$
$$Y_2 = w_2 + w_1' y_1 + w_1 y_2$$
$$z = y_2$$

An examination of Y_1 shows all prime implicants are covered. However for Y_2 there is an additional prime implicant (Y_2 Y_1) which is not grouped. We need to include this term else there will be static hazards.



But, should check for dynamic hazards first, anyway

Circuit of Q1 reduces to the excitation diagram below (from previous tut)

Present State (y_2y_1)	w_2 w_1 00	01	11	10	Output z
(A) 00	**00**	01	10	10	0
(B) 01	11	**01**	10	10	0
(C) 11	**11**	**11**	10	**11**	1
(D) 10	00	11	**10**	**10**	1

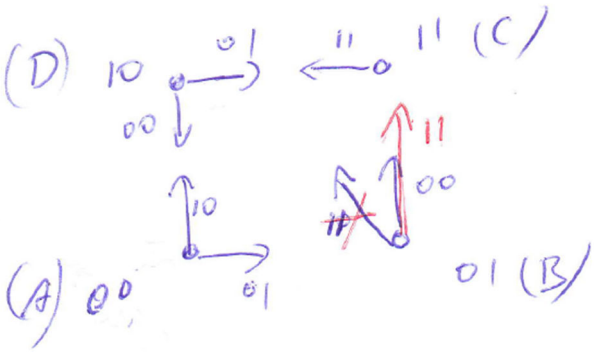
But state A (00) and state B (01) have a single state. Must be don't cares.

Present State (y_2y_1)	w_2 w_1 00	01	11	10	Output z
(A) 00	**00**	01	xx	10	0
(B) 01	11	**01**	10	xx	0
(C) 11	**11**	**11**	10	**11**	1
(D) 10	00	11	**10**	**10**	1

Upon drawing the transition diagram we see

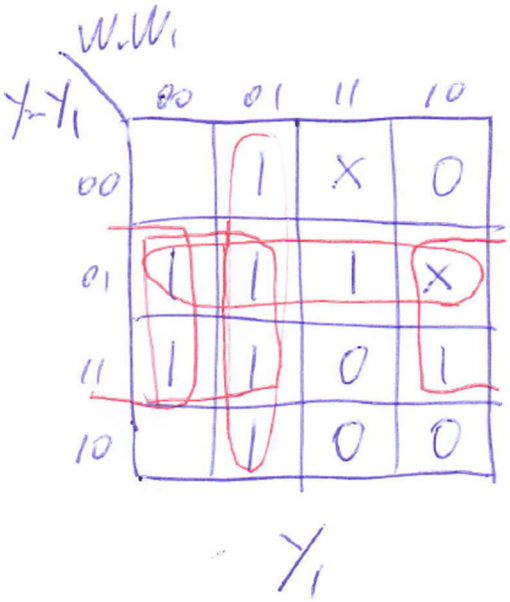
That there is a dynamic hazard.

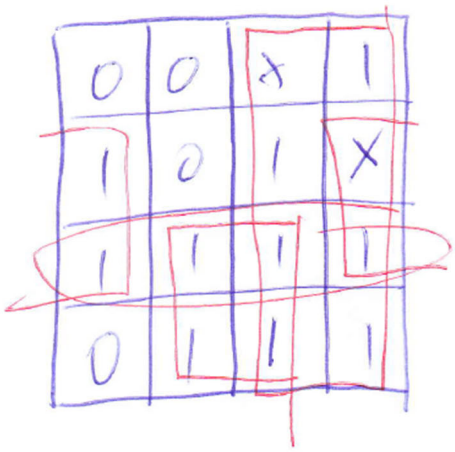
This can be eliminated by forcing state 01 to transition through state 11. Note: Could also go through state 00.



Present State (y_2y_1)	w_2 w_1 00	01	11	10	Output z
(A) 00	**00**	01	xx	10	0
(B) 01	11	**01**	11	xx	0
(C) 11	**11**	**11**	10	**11**	1
(D) 10	00	11	**10**	**10**	1

Y2 is the same (after we include the extra essential prime implicant)





But Y1 has changed. The equations are:

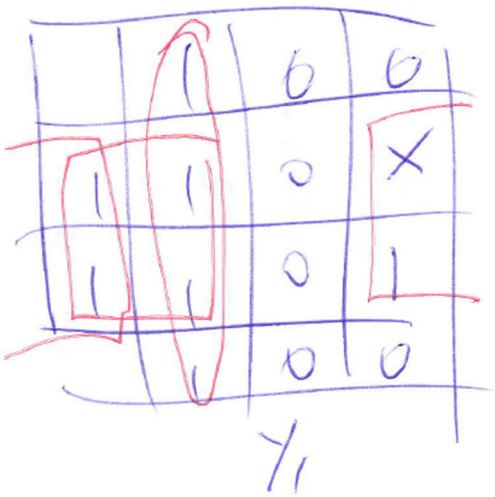
$Y_1 = w_1 w_2' + w_2' y_1 + y_1 y_2' + Y_1 W_1'$

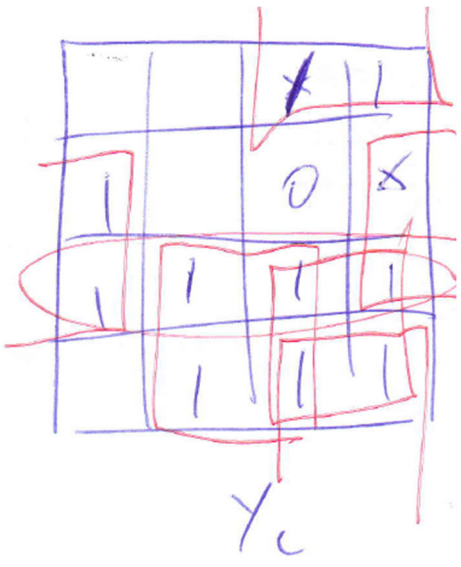
$Y_2 = Y_2 Y_1 + Y_2 W_1 + W_2 + Y_1 W_1$

$z = Y_2$

You were not asked to redraw the circuit diagram.

Note: If you changed state B to go through state A you will get a different answer, below.

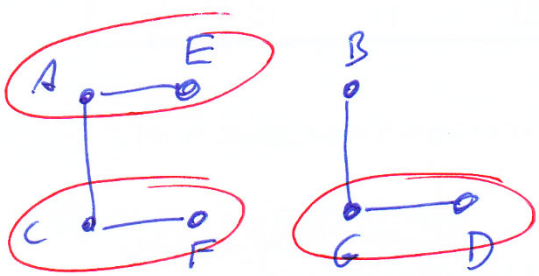




Q2: Reduce the number of states. First partition: P_1 = (ACEF)(BDG)

Present State	w_2 w_1 00	01	10	11	Output z
A	**A**	E	C	--	0
B	--	E	G	**B**	1
C	A	--	**C**	F	0
D	A	**D**	--	B	1
E	A	**E**	--	B	0
F	--	D	C	**F**	0
G	A	--	**G**	B	1

The second partition is P_2 = ((AC)(AE)(CF))((BG)(DG))
It is fairly obvious from the merger diagram that we merge states (AE) and (CF), but not (AC); and states (DG) (or (BG).



So the reduced flow table is:

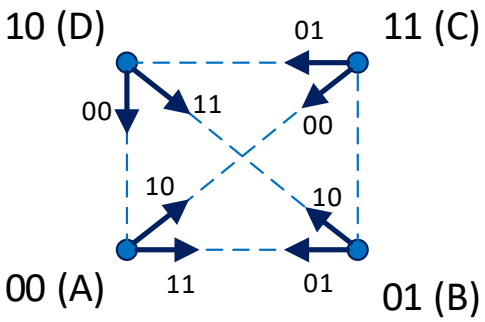
Present State	w_2 w_1 00	01	10	11	Output z
A	**A**	**A**	C	B	0
B	--	A	D	**B**	1
C	A	D	**C**	**C**	0
D	A	**D**	**D**	B	1

Now we should put the flow table into the form with grey code inputs.

Present State	w_2 w_1 00	01	11	10	Output z
A	**A**	**A**	B	C	0
B	--	A	**B**	D	1
C	A	D	**C**	**C**	0
D	A	**D**	B	**D**	1

Firstly, lets make a state assignment, A = 00; B = 01; C = 11; and D = 10, Then we can draw the state transition diagram below

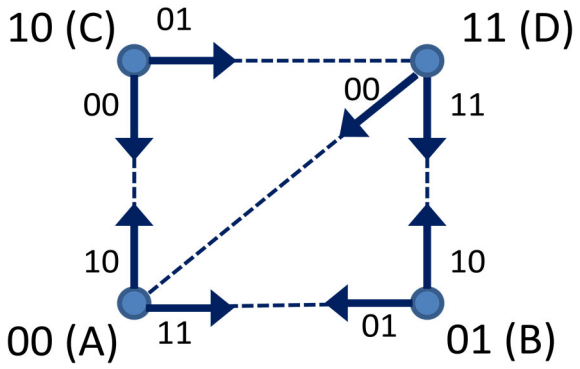
Clearly we can see many diagonal transitions. This indicates a poor state assignment, so try another.



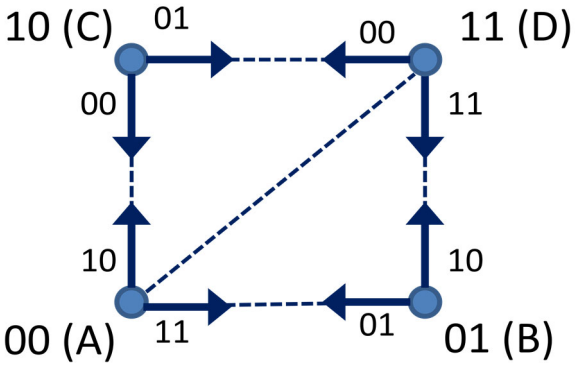
If we make a state assignment, A = 00; B = 11; C = 10; and D = 01, we still get multiple diagonals.

But if we make a state assignment, A = 00; B = 01; C = 10; and D = 11, we get the state transition diagram (with 1 diagonal) below

Present State	$w_2 w_1$ 00	01	11	10	Output z
A	**A**	**A**	B	C	0
B	--	A	**B**	D	1
C	A	D	**C**	**C**	0
D	A	**D**	B	**D**	1



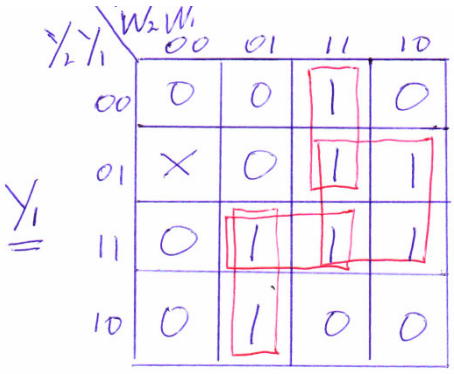
This is a much better state assignment as there is only one diagonal transition (from D to A which could cause a race condition) that we need to deal with. In this instance it is simple to fix as we can go via state C. That is for state D with input {00}, move first to C, which for input {00} will just transition to state A

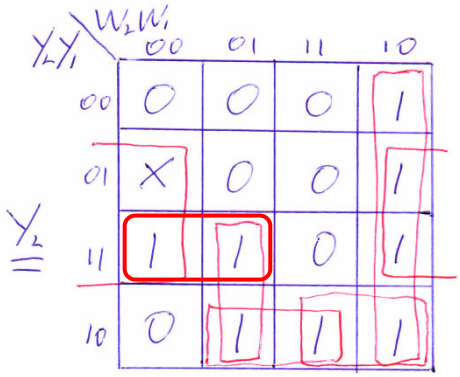


Hence, the new excitation table is:

Present State Y_2Y_1	$w_2 w_1$ 00 Y_2Y_1	01 Y_2Y_1	11 Y_2Y_1	10 Y_2Y_1	Output z
00	**00**	**00**	01	10	0
01	--	00	**01**	11	1
11	10	**11**	01	**11**	1
10	00	11	**10**	**10**	0

Then we can get the K-map for Y_1 (top) and Y_2 (bottom) as:





This leads to the next state and output expressions:

$Y_1 = w_1y_1 + w_2w_1y_2' + w_1w_2'y_2 + w_1y_1y_2$

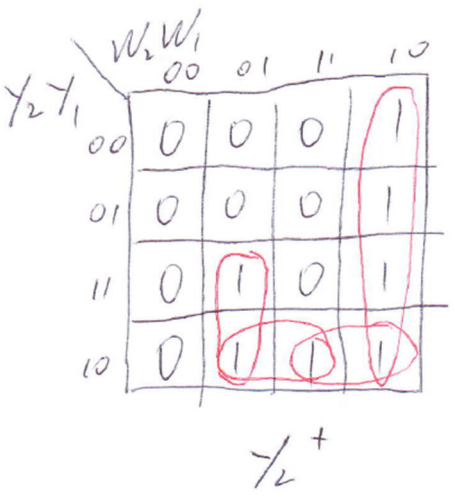
$Y_2 = w_2w_1' + y_1w_1' + w_2y_2y_1' + w_1y_2y_1' + w_1w_2'y_2 + w_2'y_2y_1$

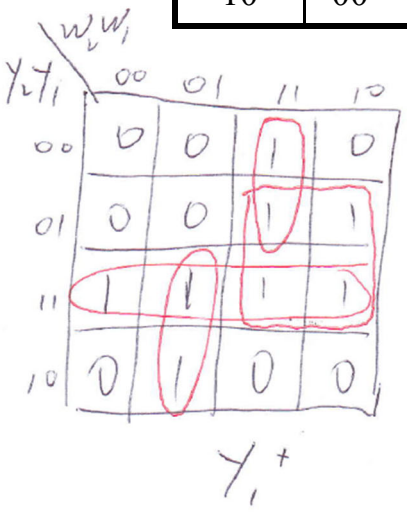
$z = y_1$

From which we can get the logic circuit (not shown)

Note: we could have also used the don't care in state B, resulting in the excitation table below

Present State Y_2Y_1	w_2 w_1 00	01	11	10	Output z
	Y_2Y_1	Y_2Y_1	Y_2Y_1	Y_2Y_1	
00	**00**	**00**	01	10	0
01	00	00	**01**	11	1
11	01	**11**	01	**11**	1
10	00	11	**10**	**10**	0





Both Y_1 and Y_2 result in simpler expressions.